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Evidence for orbital superfluidity in the *P*-band of a bipartite optical square lattice

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The successful emulation of the Hubbard model in optical lattices has stimulated extensive efforts to extend their scope to also capture more complex, incompletely understood scenarios of many-body physics. A promising approach is to consider higher bands, where the orbital degree of freedom gives rise to a structural diversity that is directly relevant, for example, for the physics of strongly correlated electronic matter. Here we report evidence for the formation of a superfluid in the *P*-band of a bipartite optical square lattice with *S*-orbits and *P*-orbits arranged in a chequerboard pattern. The observed momentum spectra feature cross-dimensional coherence with a lifetime of nearly 20 ms. Depending on the value of a small adjustable anisotropy of the lattice, our findings are explained either by real-valued striped superfluid order parameters with different orientations $P_x \pm P_y$, or by a complex-valued $P_x \pm iP_y$ order parameter, which breaks time-reversal symmetry.

Optical lattices are quantum degenerate gases trapped in synthetic lattices formed by the interference of multiple laser beams. In their ground state they represent nearly ideal emulations of single-flavour Hubbard models[1–5]. According to a general consideration by Feynman (known as the 'no-node' theorem[6,7]), bosons under very general circumstances exhibit a positive definite ground state, which limits their usefulness to simulate relevant many-body physics. A promising way around Feynman's statement is to consider higher bands in optical lattices. Such orbital optical lattices should allow the preparation of exciting new many-body scenarios with relevant counterparts in the area of strongly correlated electronic matter, where the orbital degree of freedom with its rich structure of possible orientations gives rise to a wealth of interesting phenomena. For example, in the transition-metal oxides orbital physics is believed to be a key element for understanding their metal–insulator transitions, superconductivity and colossal magnetoresistance. Recent theoretical proposals have considered, for example, the formation of multiflavour and multiorbital systems[8–11], supersolid quantum phases in cubic lattices[12,13], quantum stripe ordering in triangular lattices[14] and Wigner crystallization in honeycomb lattices[15]. The recent prediction, that the lifetime of atoms in higher Bloch bands can be unexpectedly long, has further strengthened the potential significance of orbital physics in optical lattices[8,9]. Previous experiments have demonstrated that the excitation of higher Bloch bands is in fact possible[16,17]. In a recent experiment, investigating a homogeneous quasi one-dimensional (1D) lattice of bosons, transient partial coherence in the *P*-band has been observed[18]. In an extension to a 2D square lattice, however, no cross-coherence could be established, which was previously predicted in ref. 8 as a consequence of the small transverse tunnelling rate of the *P*-orbitals.

Here, we explore orbital superfluidity in a bipartite square optical lattice. We report evidence for the first observation of a 2D orbital superfluid in the *P*-band. A key feature responsible for the formation of full cross-dimensional coherence is the composition of the lattice from two types of site with different well depths arranged as the black and white domains of a chequerboard. The deeper wells provide local *P*-orbits, which mainly support longitudinal tunnelling; the shallow wells contribute isotropic *S*-orbits, which act as cross-dimensional tunnelling links. By means of a population swapping technique we can selectively excite a large fraction of the atoms into the lowest *P*-band. Although initially the entire second Brillouin zone is smoothly populated and the sample does not show any coherence, after a few milliseconds the atoms accumulate at the local minima of the *P*-band energy surface and coherence with respect to both lattice dimensions arises. Only after times of the order of several tens of milliseconds is coherence lost as the atoms undergo collisions repopulating the ground state *S*-band. Depending on the adjustment of a small anisotropy in our lattice potential, which breaks the four-fold rotational symmetry of the *P*-band, our observations are explained by the formation of striped (finite momentum) superfluid orders with different orientations or, if isotropy of the *P*-band is established, a genuinely complex-valued superfluid order, which exhibits staggered orbital currents. This latter case breaks time-reversal symmetry, indicated by the presence of a sheet of staggered vortices with a unit flux quantum per plaquette. It realizes a variant of the π-flux model[19] proposed in connection with cuprate superconductors and represents a striking example beyond Feynman's no-node theorem[6,7].

Lattice potential with tunable anisotropy

The core of our experimental set-up is a (quasi 2D) square optical lattice, composed of two classes of (tube-shaped) lattice sites (denoted as $\mathcal{A}$ and $\mathcal{B}$) arranged as shown in Fig. 1a. The accordant optical potential is formed by crossing two optical standing waves oriented along the x and y axes with linear polarizations parallel to the z axis, produced in the two branches of a Michelson interferometer (details are deferred to the Methods section). This allows us to adjust the time phase difference between the standing waves θ to any desired value[20]. For $\theta < \pi/2$ the $\mathcal{A}$-sites are shallower than the $\mathcal{B}$-sites and vice versa. Two different kinds of anisotropy can occur in the optical potential. The first type, which affects the structure of the energy minima in the *P*-band and hence the possible superfluid order, results from the fact that because of imperfect reflection optics the intensity ratios between the two travelling waves forming each of the two standing waves are generally unequal. To control this anisotropy, appropriate polarization optics in one of the standing waves allows us to rotate the linear polarization of the incident beam away from the z axis by a small angle α while maintaining the polarization of the reflected beam parallel to the

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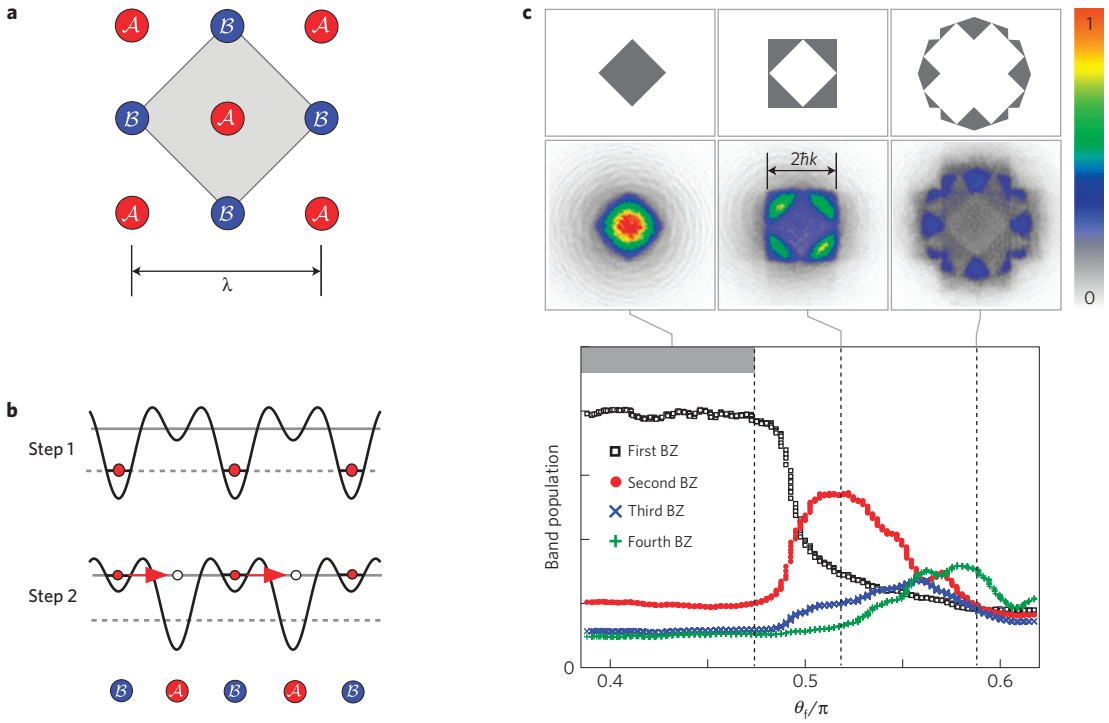


Figure 1 | Population of excited bands. a, The lattice comprises two classes of lattice sites denoted by $\mathcal{A}$ and $\mathcal{B}$. The grey area denotes the Wigner–Seitz unit cell of the $\mathcal{A}$-sublattice. **b**, Experimental sequence used to populate excited bands. **c**, The populations within the first four Brillouin zones (BZ) are plotted (according to the colour scale shown) versus the final value θ_f adjusted in step 2 of the sequence in **b**. The upper panel shows illustrations of the first, second and fourth Brillouin zones (top row) and their observed populations for values of θ_f within the grey bar and at the dashed lines (bottom row).

z axis. Tuning of α allows us to adjust the structure of the energy minima in the P-band, which are degenerate if $\alpha = \alpha_{iso} \approx \pi/5$. For $\alpha < \alpha_{iso}$ or $\alpha > \alpha_{iso}$ this degeneracy is lifted, and hence two inequivalent minima with different energies arise. A second kind of anisotropy results from an imbalance of the intensities in both interferometer axes due to imperfect beam splitting optics. As a result of using finite-sized Gaussian beams, a local imbalance of the standing-wave intensities is unavoidable even if perfect beam splitting optics was available. If $\alpha \approx \alpha_{iso}$, this anisotropy does not notably affect the structure of the P-band energy minima, where condensation takes place, and thus it is irrelevant to the structure of the superfluid order possible in our experiment. For details, see the Methods section.

Population of higher bands

We may populate excited bands through the following two-step population swapping procedure related to the techniques used in ref. 21 (Fig. 1b). Initially a Bose–Einstein condensate (BEC) of rubidium (^{87}Rb) atoms is prepared and the lattice potential is activated with $\theta = \theta_i \approx 0.38\pi$, such that the well depth of the $\mathcal{B}$-sites is about 4.5 times larger than that of the $\mathcal{A}$-sites (Methods section). A ground-state lattice is thus formed with the shallow $\mathcal{A}$-wells contributing little to the tunnelling dynamics. In the second step, θ is rapidly increased (within 0.2 ms, which is shorter than the nearest-neighbour tunnelling time) to a final value $\theta = \theta_f > \pi/2$ such that the $\mathcal{A}$-wells become significantly deeper than the $\mathcal{B}$-wells (Fig. 1b). Depending on the chosen value of θ_f, population is transferred to excited bands according to the graphs shown in Fig. 1c. As expected, excitations to higher bands are not observed for $\theta_f < \pi/2$. In this domain the θ-independent background population in the excited bands is a consequence of imperfections in our adiabatic band mapping procedure and a background of uncondensed atoms arising outside the image plane because of a technical peculiarity of our BEC production[22]. For suitable values of $\theta_f > \pi/2$ the second (lower P-band) and the fourth (lowest D-band) bands become the most populated. Notably, the third band (upper P-band) population always remains small.

Observations

For the observations reported here, the population ratio between the second and the first band is maximized ($\theta_f = 0.53\pi$). After a variable hold time, we map the populations in the different Brillouin zones or record the momentum distribution of the atoms. We first studied the case $\alpha = 0$, when the four-fold rotational symmetry of the P-band is broken. In Fig. 2a the distribution of the atomic population among the different Brillouin zones is shown for hold times between 0.1 and 200 ms. After 0.1 ms the second Brillouin zone is nearly uniformly populated. As time evolves, sharp peaks arise at the two finite quasi-momenta $\mathbf{K}_{(-,+)}$ and $\mathbf{K}_{(+,-)}$ (located on the boundary between the first and the second Brillouin zone), developing maximal contrast at hold times around 12 ms and persisting for times up to 40 ms. Here, $\mathbf{K}_{(\pm,\pm)} \equiv 1/2(\pm\hbar k, \pm\hbar k)$, where $k \equiv 2\pi/\lambda$ and λ denotes the wavelength of the lattice beams. As the hold time is increased beyond a few tens of milliseconds the sharp peaks disappear and the first Brillouin zone is repopulated. In the corresponding momentum spectra (Fig. 2b) beyond 2 ms hold time, pronounced maxima arise at the momenta $\mathbf{K}_{(-,+)}$ and $\mathbf{K}_{(+,-)}$, accompanied by higher-order Bragg peaks (highlighted for the 12 ms picture by dashed red circles in Fig. 2b), which indicate the emergence of cross-dimensional coherence. The width of the observed Bragg peaks, which is limited by interaction effects, insufficient expansion times and technical imperfections of the imaging, sets a lower bound of the coherence area of 5×5 $\mathcal{A}$-sites. When we carefully tune α to adjust the isotropy of the P-band (the value of α has to match α_{iso} to better than $\delta\alpha \approx \pm\pi/100$ (Methods section)), the atoms are observed to accumulate at the four momenta $\mathbf{K}_{(-,+)}$, $\mathbf{K}_{(+,-)}$, $\mathbf{K}_{(+,+)}$ and $\mathbf{K}_{(-,-)}$ in the corresponding Brillouin zone mappings and momentum

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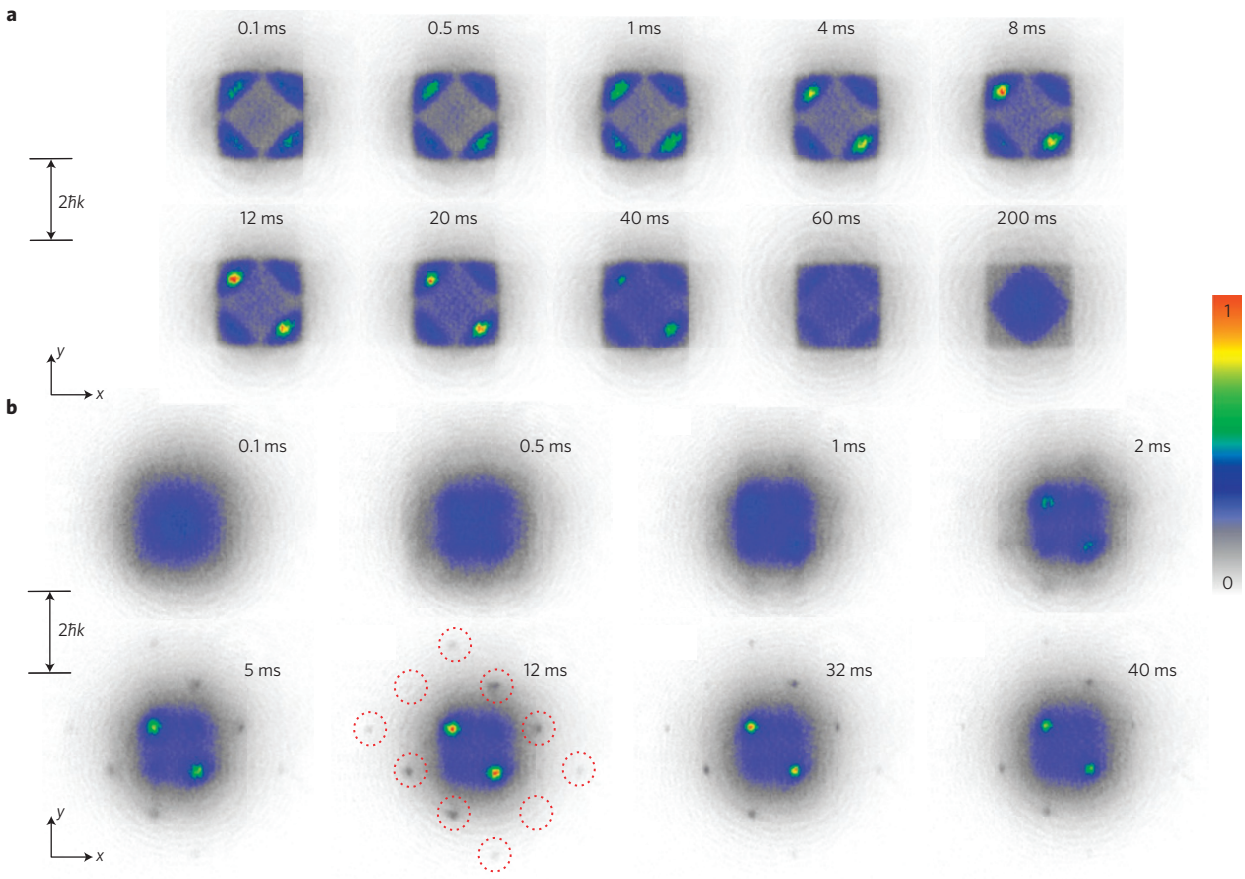


Figure 2 | Evolution of band populations and momentum distribution. a, Mapping of band populations for increasing hold times in the lattice. **b**, Corresponding momentum spectra. The red dashed circles in the 12 ms spectrum highlight the higher-order Bragg peaks. The lattice axes are oriented in the x and y directions. The case $\alpha = 0$ is shown. Relative particle numbers are parameterized by the colour scale shown.

spectra (Fig. 3a). As shown in Fig. 3b, we may invert the initial anisotropy by adjusting $\alpha > \alpha_{\rm iso} + \delta\alpha$ and recover an accumulation at two momenta, however with an orientation rotated by $\pi/2$ with respect to the observations for $\alpha < \alpha_{\rm iso} - \delta\alpha$. In Fig. 3c–g the 12 ms momentum spectrum of Fig. 2b (replotted in Fig. 3c) is analysed more quantitatively. In Fig. 3d a side view of this spectrum along the direction denoted by S_3 in Fig. 3c is shown. In Fig. 3e a sectional view along the direction denoted by S_1 in Fig. 3c is shown. The solid grey line, matching well with the observed data (black circles), shows a fit to a sum of the (tight binding) momentum Wannier functions of the S- and P-bands, to account for the incoherent fractions in both bands, and of two equal Gaussian functions separated by $\sqrt{2}\hbar k$, to account for the coherent fraction. The widths of the S-band Wannier function and the double Gaussian function, as well as the individual amplitudes of the three functions, are used as fit parameters. The contributions of the P- and S-band Wannier functions and the double Gaussian function are shown by the dashed red, dotted blue and solid green curves, respectively. The sectional view in Fig. 3f is along the direction denoted by S_2 in Fig. 3c. The grey line repeats the incoherent fraction obtained from the data in Fig. 3e. In Fig. 3g, a section along the direction denoted S_4 in Fig. 3c is shown. Two Gaussian functions are used to fit the data, yielding the coherent and incoherent fractions given by the solid green and dotted blue curves. Figure 3e–g clearly shows that a substantial fraction of the atoms are coherent.

Topology of orbital order

We interpret the accumulation of atoms at specific quasi-momenta as being due to a collision-aided condensation process transferring energy from the two dimensions of the lattice into the third weakly confined dimension. The two inequivalent pairs of equivalent quasi-momenta $\mathbf{K}_{(\pm,\pm)}$ and $\mathbf{K}_{(\mp,\pm)}$ (indicated by red and blue discs in Fig. 4a), where atoms are observed to condense, correspond to the two inequivalent local energy minima of the lowest P-band (Fig. 5). If $\alpha = \alpha_{\rm iso}$, these minima are degenerate and condensation at both points $\mathbf{K}_{(+,-)}$ and $\mathbf{K}_{(+,+)}$ is possible. If $\alpha < \alpha_{\rm iso}$ or $\alpha > \alpha_{\rm iso}$, the degeneracy is lifted and condensation selects the energetically lower lying minimum. Similar condensation timescales are found for all values of α. Our observations show that for sufficiently weak interactions the quantum state prepared by the condensation process is a coherent superposition or an incoherent mixture of the two Bloch states $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ and $\phi_{\mathbf{K}_{(+,+)}}(x,y)$ corresponding to the two possible condensation points. These Bloch states are derived from a band calculation for the general potential in equation (1) of the Methods section including the lowest nine bands. Assuming a coherent superposition $\psi = a\phi_{\mathbf{K}_{(+,-)}}(x,y) + b\phi_{\mathbf{K}_{(+,+)}}(x,y)$ with complex α-dependent coefficients a and b (the possibility of an incoherent mixture is discussed below), we may readily calculate the corresponding momentum spectrum. We find that, to obtain agreement with the observed spectrum, the possible choice of the coefficients a and b is constrained by the relations $b = \pm ia$, if $\alpha = \alpha_{\rm iso}$, by $b = 0$, if $\alpha < \alpha_{\rm iso}$, and by $a = 0$, if $\alpha > \alpha_{\rm iso}$. For example, for $\alpha = \alpha_{\rm iso}$ the calculated momentum spectrum (with a width of the Bragg peaks according to a lattice with 5×5 $\mathcal{A}$-sites) is depicted in Fig. 4b, showing good agreement with the corresponding experimental counterpart in Fig. 3a.

The topology of the wavefunction ψ exhibits a chequerboard pattern of local S-orbits in the shallow wells and local P-orbits in

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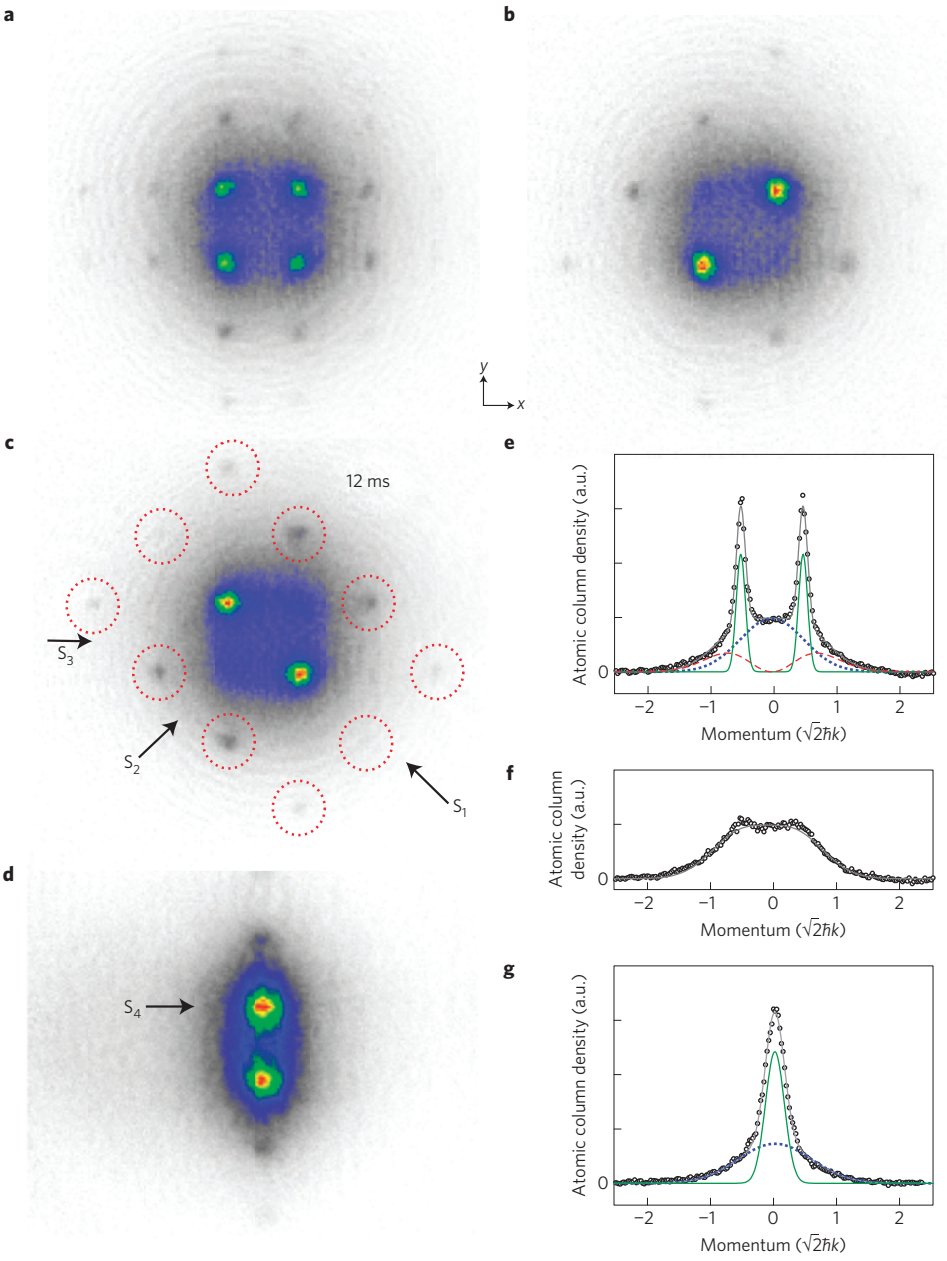


Figure 3 | Tuning of anisotropy and condensate fractions. a, Momentum spectrum for an isotropic *P*-band $\alpha = \alpha_{iso}$. **b**, Momentum spectrum for the case of $\alpha > \alpha_{iso}$. **c**, The 12-ms graph of Fig. 2b is replotted with the red dashed circles highlighting the higher order Bragg peaks. **d**, Side view of the momentum spectrum in **c** along the direction S_3. **e**, Sectional view along the direction S_1 in **c**. **f**, Sectional view along the direction S_2 in **c**. **g**, Sectional view along the direction S_4 in **d**. The solid grey lines in **e**,**f** and **g** are fits to the sum of the (tight-binding) momentum Wannier functions of the *S*- (dotted blue trace) and *P*-bands (dashed red trace) accounting for the incoherent fraction and Gauss-functions (solid green trace) for the coherent fraction.

the deep wells, where the local *P*-orbits are $P_x - P_y$, if $\alpha < \alpha_{iso}$, $P_x + P_y$, if $\alpha > \alpha_{iso}$, and alternating $P_x \pm iP_y$, if $\alpha = \alpha_{iso}$. The cases $\alpha = \alpha_{iso}$ and $\alpha < \alpha_{iso}$ are illustrated in Fig. 4c,d. A $2\lambda \times 2\lambda$ section of the lattice is shown with the deep $\mathcal{A}$-sites and the shallow $\mathcal{B}$-sites indicated by large and small grey discs respectively. The plotted numbers denote the local phases of the wavefunction. The physics of ψ can be understood by extending the arguments discussed in refs 8,9. In the tight-binding limit the wavefunction near a lattice site resembles a harmonic oscillator state. The population swapping sequence provides a local *S*-state at the shallow $\mathcal{B}$-wells. The possible states at the deeper $\mathcal{A}$-sites are superpositions of local P_x and P_y orbits, which in the absence of collisional interactions and for $\alpha = \alpha_{iso}$ are energetically degenerate. The repulsive interaction favours a maximum angular momentum at these sites with an orbit configuration $P_x \pm iP_y$, corresponding to angular momentum quanta $\pm\hbar$ per atom[8,9]. To maximize intersite hopping, the local phases of adjacent orbits match at their tunnelling junction. The corresponding wavefunction (Fig. 4c) has remarkable properties. With respect to the $\mathcal{A}$-sites it exhibits a pattern of staggered orbital angular momenta. With respect to the $\mathcal{B}$-sites it exhibits vortical plaquette currents and thus resembles the π-flux phase discussed in the context of high-T_c superconductors[19] and the staggered vortex superfluid, recently discussed in refs 23–25. Away from $\alpha = \alpha_{iso}$ a difference in the single-particle energies of the $P_x \pm P_y$-orbits is introduced that may exceed the collisional energy gain of the maximum angular momentum orbits. A band calculation for the case $\alpha = 0$ (and $\theta = 0.53\pi$) confirms that the energies of the band minima at $\mathbf{K}_{(+,-)}$ and $\mathbf{K}_{(-,+)}$ are lowered with respect to those at

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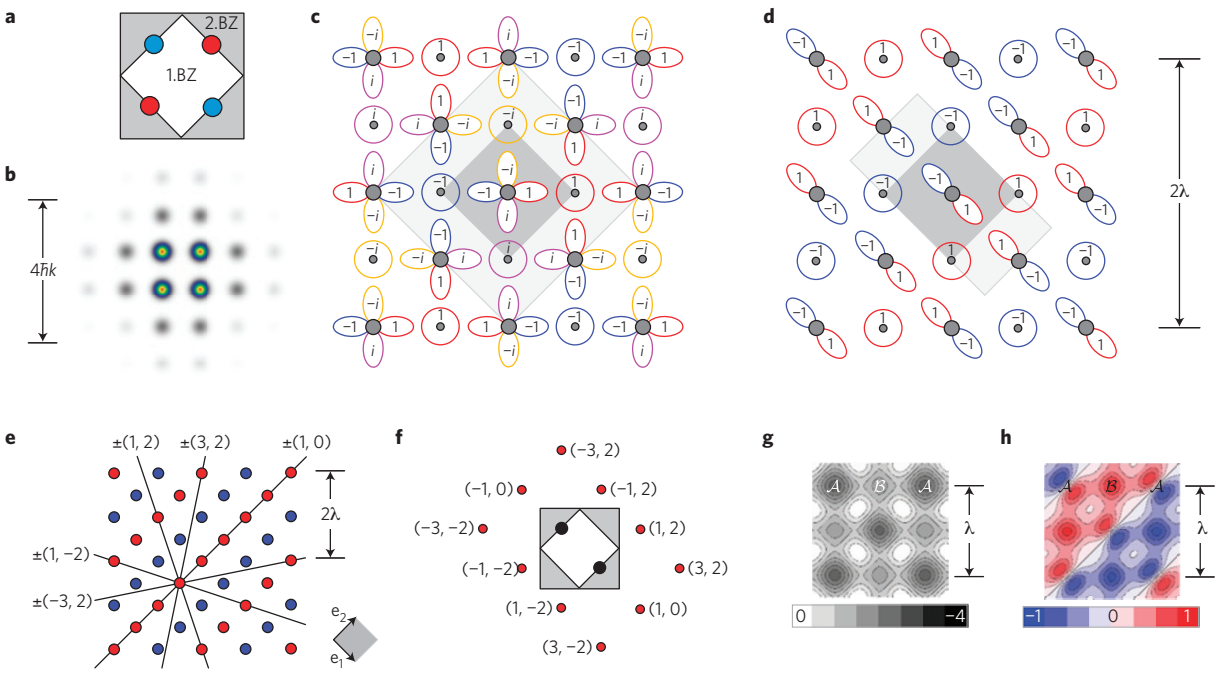


Figure 4 | Nature of superfluid order. a, The first (white area) and second (grey area) Brillouin zones (BZ) and the quasi-momenta corresponding the two possible inequivalent energy minima of the *P*-band (red and blue discs). **b**, Momentum spectrum calculated for $\alpha = \alpha_{iso}$. **c**, Orbital ordering for $\alpha = \alpha_{iso}$. **d**, Orbital alignment in the case where the lattice potential exhibits a small anisotropy ($\alpha < \alpha_{iso}$). The dark (light) grey areas in **c** and **d** show the unit cell of the lattice potential (order parameters). **e**, Schematic representation of the phase ordering in **d**. The black lines show the orientation of plane wavefronts (labelled with their Miller indices with respect to the primitive vectors $\mathbf{e}_1, \mathbf{e}_2$) compatible with the imposed phase order. **f**, Positions of the Bragg resonances (small red discs) in momentum space corresponding to the wavefronts in **e**. **g**, Lattice potential for $\theta = 0.56\pi$, $\alpha = 0$ and a well depth parameter $\tilde{V}_0 = 6.2E_{rec}$. **h**, Bloch function $\phi_{\mathbf{K}_{(+,-)}}(x, y)$ for the potential in **g** corresponding to the quasi-momentum $\mathbf{K}_{(+,-)}$ in **a** (blue discs).

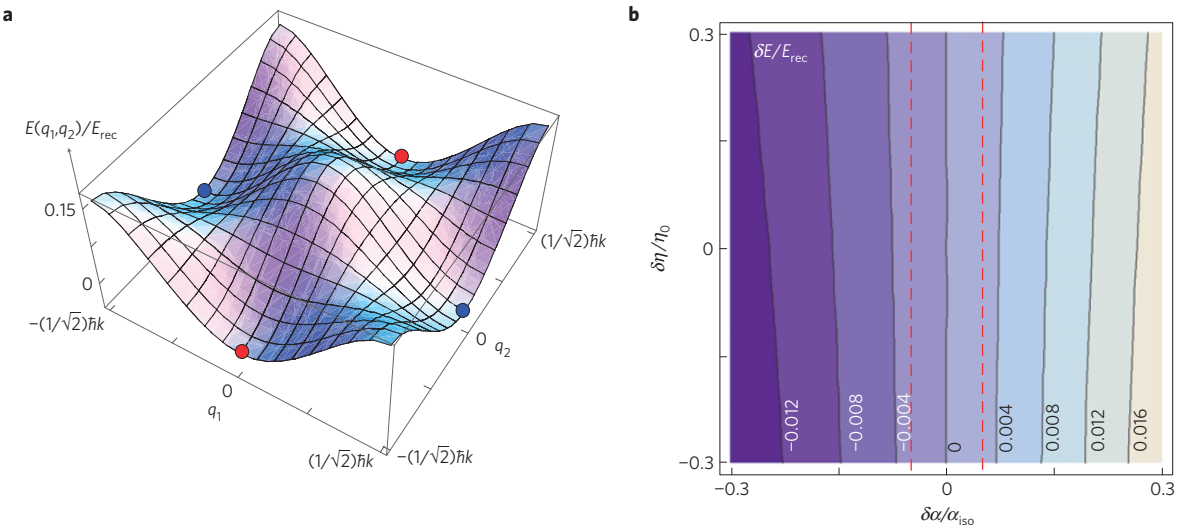


Figure 5 | Anisotropy and *P*-band energy minima. a, Energy surface $E(q_1, q_2)$ of the *P*-band plotted within the first Brillouin zone for $\alpha = \alpha_{iso}$. The red and blue discs indicate the local minima. Discs indicating equivalent minima share the same colour. **b**, The energy difference δE of the two inequivalent local energy minima is plotted versus $\delta\alpha$ and $\delta\eta$.

$\mathbf{K}_{(+,+)}$ and $\mathbf{K}_{(-,-)}$ by a few thousandths of the atomic recoil energy $E_{rec} \equiv \hbar^2 k^2 / 2m$ (m denotes the atomic mass) or a few per cent of the energy width of the band (Methods section), which corresponds to sub-nanokelvin temperatures. At the $\mathcal{A}$-sites, instead of local $(P_x \pm iP_y)$-orbits, $(P_x \pm P_y)$-orbits are then energetically favourable and a striped order with zero local angular momentum arises (the case $\alpha < \alpha_{iso}$ being sketched in Fig. 4d). Remarkably, both wavefunctions in Fig. 4c,d break the translational symmetry of the lattice potential, as is seen from their different unit cells plotted in the figure.

Higher-order Bragg peaks, role of local *S*-orbits

The wavefunctions of Fig. 4c,d readily explain the higher-order Bragg peaks observed in the corresponding momentum spectra. In Fig. 4e, the $\mathcal{B}$-sublattice is replotted for the case of the striped superfluid order (Fig. 4d), with the red and blue discs denoting sites with opposite local phase. The black lines show the wavefronts of possible plane waves that can propagate through the lattice without suffering destructive interference due to the imposed phase order. These planes, which are the lattice planes comprising only

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$\mathcal{B}$-sites with the same local phase, are denoted by their Miller indices. In Fig. 4f, the expected Bragg resonances in momentum space are plotted and in agreement with the observations in Fig. 2b. Similar arguments apply for the case of the staggered angular momentum order in Fig. 4c.

It is instructive to observe the important role of the local S-orbits in the shallow $\mathcal{B}$-wells. For simplicity, we consider the striped state. As is seen from Fig. 4d, the S-orbits are essential for establishing coherence perpendicular to the direction of alignment of the $(P_x - P_y)$-orbits. Transverse tunnelling between P-orbits is typically an order of magnitude weaker than axial tunnelling[8,9]. Tranverse tunnelling between $\mathcal{A}$-wells is thus expected to occur predominantly via the $\mathcal{B}$-wells. This picture is well confirmed by plotting the lattice potential (Fig. 4g) and the corresponding Bloch function $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ (Fig. 4h). For better visibility of the P-orbits we have used $\theta = 0.56\pi$ here. An analogous consideration applies for the staggered angular momentum order in Fig. 4c. In fact, if resonant tunnelling between $\mathcal{A}$- and $\mathcal{B}$-wells is suppressed by adiabatically switching off the $\mathcal{B}$-wells after the superfluid has been established, a rapid loss of transverse coherence is observed in the momentum spectra. Furthermore, a significant reduction of the lifetime of atoms in the P-band is observed. The latter finding is understood by noting that collisions within the $\mathcal{B}$-wells, where the atoms are in their local ground state, cannot reduce the P-band population. If, however, the atoms are constrained to reside exclusively in the $\mathcal{A}$-wells, enhanced collisional decay to the lowest band is expected. Long P-band lifetimes in a bipartite lattice have previously been predicted[26].

Complex order versus fragmentation

For the anisotropic cases $\alpha < \alpha_{\mathrm{iso}}$ or $\alpha > \alpha_{\mathrm{iso}}$ the observed momentum spectra (Fig. 3b,c) uniquely determine the wavefunctions to be either $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ or $\phi_{\mathbf{K}_{(+,+)}}(x,y)$. The isotropic case, however, requires a more careful discussion. As a calculation confirms, an incoherent mixture of the striped orders $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ and $\phi_{\mathbf{K}_{(+,+)}}(x,y)$ with equal weights would lead to a very similar momentum spectrum to that of Fig. 3a. In principle two possibilities are to be considered: the first scenario, where the two incoherent components coexist at the same location, and a fragmentation scenario, where the lattice decomposes into disjunct domains exhibiting striped orders with orthogonal orientations. The first scenario seems fairly unlikely, as both states $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ and $\phi_{\mathbf{K}_{(+,+)}}(x,y)$ share common local S-orbits, such that a precise determination, whether a particle belongs to one or the other phase, which would result in an indeterminate relative phase, seems difficult to conceive. Two kinds of fragmentation scenario may be considered, caused by either an uncontrolled local variation of the balance between the incident and reflected intensities in the crossed standing waves, or by the weak magnetic trap potential present in our experiments. The former case can be excluded. The intensity imbalance that has to be introduced by means of polarization adjustment (that is, tuning α) to switch from the isotropic to the anisotropic case is approximately five per cent. Careful beam profile measurements and the 99.5% interference contrast obtained in our Michelson interferometer set-up show that over a small distance of 20 μm, which corresponds to the sample diameter, the local variation of the intensity mismatch between the laser beams forming the standing waves is far beyond this level. We also do not expect that the weakly confining (isotropic) harmonic trap potential (40 Hz trap frequency in each direction) of the magnetic trap could give rise to fragmentation because the two possible fragments in question (described by wavefunctions $\phi_{\mathbf{K}_{(+,-)}}(x,y)$ and $\phi_{\mathbf{K}_{(+,+)}}(x,y)$, which are congruent up to a $\pi/2$ rotation) do not differ with respect to their individual chemical potentials. This is in contrast to the observed spatial separation of paired and unpaired fermions in optically trapped unbalanced fermionic spin mixtures[27,28], which may be understood as a consequence of the different chemical potentials of paired and unpaired fermions[29]. In conclusion, we claim evidence that the momentum spectrum observed in Fig. 3a in fact originates from the order parameter of Fig. 4a. A stringent proof, however, remains an important challenge for future theoretical and experimental efforts. On the experimental side, this could be possibly achieved by means of detection schemes operating in configuration space with single-site resolution[30].

Methods

Optical potential. We couple a laser beam at a wavelength $\lambda = 1{,}064$ nm and $1/e^2$ radius $w_0 = 100$ μm to a Michelson interferometer with its two branches crossed at a right angle to obtain the optical potential

$$-\frac{\bar{V}_0}{4}\mathrm{e}^{-\frac{2z^2}{w_0^2}}\left|\eta[(\hat{z}\cos(\alpha)+\hat{y}\sin(\alpha))\mathrm{e}^{ikx}+\epsilon\hat{z}\mathrm{e}^{-ikx}]+\mathrm{e}^{i\theta}\hat{z}(\mathrm{e}^{iky}+\epsilon\mathrm{e}^{-iky})\right|^2 \qquad (1)$$

where η accounts for a small difference in the powers directed to both interferometer branches (a typical value for η is $\eta_0 \approx 0.95$), $\epsilon \approx 0.81$ accounts for the power reduction in the retro-reflected beams caused by imperfect optics, and the angle α permits us to adjust the degree of anisotropy introduced if $\epsilon \neq 1$. $\hat{x},\hat{y},\hat{z}$ are the unit vectors in the respective directions. An isotropic P-band arises if $\cos(\alpha) \approx \epsilon$, which amounts to $\alpha = \alpha_{\mathrm{iso}} \approx \pi/5$. The four local energy minima arising in the P-band at the edge of the first Brillouin zone (Fig. 5a) are then degenerate. Tuning of α and η affects the energy minima of the P-band in a different way. Remarkably, if $\alpha = \alpha_{\mathrm{iso}}$ is adjusted, the structure of the P-band energy minima is not at all modified by tuning η. This is shown in Fig. 5b, where the energy difference δE of the two inequivalent local energy minima of the P-band (red and blue discs in Fig. 5a) is plotted versus $\delta\alpha \equiv (\alpha - \alpha_{\mathrm{iso}})/\alpha_{\mathrm{iso}}$ and $\delta\eta \equiv (\eta - \eta_0)/\eta_0$. Within the dashed (red) lines we observe the momentum spectrum of Fig. 3a and on the left and right the spectra of Fig. 3b and c are found. The insensitivity of the P-band minima to changes of η for $\delta\alpha = 0$ is a highly relevant feature, as locally an imbalance of the standing-wave intensities is unavoidable because of the use of finite-sized Gaussian beams. The time-phase difference θ is proportional to the path length difference of the two interferometer branches, which is actively stabilized with the help of a weak additional laser beam to better than $\pi/200$. A change of θ of $\pi/4$ can be obtained in less than 0.2 ms.

Loading of lattice. A BEC of typically 2×10^5 ^{87}Rb atoms (in the $F = 2, m_F = 2$ state) is produced in a magnetic trap with standard procedures at the crossing point of the interferometer branches. The magnetic trap is adiabatically deformed to prepare a nearly isotropic condensate fraction with 12 μm radius. Within 80 ms the lattice beam intensity is ramped up with a cubic spline to $\bar{V}_0/E_{\mathrm{rec}} = 6.2$, with θ adjusted to 0.38π. This leads to depths of the $\mathcal{A}$-wells and $\mathcal{B}$-wells of $2.5E_{\mathrm{rec}}$ and $11.4E_{\mathrm{rec}}$, respectively. Excitation of the P-band is obtained by ramping θ within 0.2 ms to a value of 0.53π, thus switching the depths of the $\mathcal{A}$-wells and $\mathcal{B}$-wells to $7.5E_{\mathrm{rec}}$ and $5.0E_{\mathrm{rec}}$, respectively.

Detection procedures. Momentum spectra are recorded by switching off the lattice beams in less than 1 μs and subsequent free expansion of the atoms lasting 30 ms before an absorption image is recorded. For Brillouin zone mapping the lattice beam intensity is exponentially lowered with a time constant of 430 μs. Subsequently, ballistic expansion lasting 30 ms and absorption imaging is applied. The images shown in Figs 1, 2 and 3 are averages over five individual measurements.

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References

1. Fisher, M. P. A., Weichman, P. B., Grinstein, G. & Fisher, D. S. Boson localization and the superfluid-insulator transition. *Phys. Rev. B* **40,** 546–570 (1989).
2. Jaksch, D., Bruder, C., Cirac, J. I., Gardiner, C. W. & Zoller, P. Cold bosonic atoms in optical lattices. *Phys. Rev. Lett.* **81,** 3108–3111 (1998).
3. Greiner, M., Mandel, O., Esslinger, T., Hänsch, T. W. & Bloch, I. Quantum phase transition from a superfluid to a Mott insulator in a gas of ultracold atoms. *Nature* **415,** 39–44 (2002).
4. Lewenstein, M. *et al.* Ultracold atomic gases in optical lattices: Mimicking condensed matter physics and beyond. *Adv. Phys.* **56,** 243–379 (2007).
5. Bloch, I., Dalibard, J. & Zwerger, W. Many-body physics with ultracold gases. *Rev. Mod. Phys.* **80,** 885–964 (2008).
6. Feynman, R. P. *Statistical Mechanics: A Set of Lectures* (Addison-Wesley, 1972).
7. Wu, C. Unconventional Bose Einstein condensations beyond the 'no node' theorem. *Mod. Phys. Lett. B* **23,** 1–24 (2009).
8. Isacsson, A. & Girvin, S. Multiflavor bosonic Hubbard models in the first excited Bloch band of an optical lattice. *Phys. Rev. A* **72,** 053604 (2005).
9. Liu, W. & Wu, C. Atomic matter of nonzero-momentum Bose–Einstein condensation and orbital current order. *Phys. Rev. A* **74,** 013607 (2006).

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10. Kuklov, A. Unconventional strongly interacting Bose–Einstein condensates in optical lattices. *Phys. Rev. Lett.* **97,** 110405 (2006).
11. Xu, C. & Fisher, M. Bond algebraic liquid phase in strongly correlated multiflavor cold atom systems. *Phys. Rev. B* **75,** 104428 (2007).
12. Scarola, V. & Das Sarma, S. Quantum phases of the extended Bose-Hubbard Hamiltonian: Possibility of a supersolid state of cold atoms in optical lattices. *Phys. Rev. Lett.* **95,** 033003 (2005).
13. Scarola, V., Demler, E. & Das Sarma, S. Searching for a supersolid in cold-atom optical lattices. *Phys. Rev. A* **73,** 051601(R) (2006).
14. Wu, C., Liu, W., Moore, J. & Das Sarma, S. Quantum stripe ordering in optical lattices. *Phys. Rev. Lett.* **97,** 190406 (2006).
15. Wu, C., Bergman, D., Balents, L. & Das Sarma, S. Flat bands and Wigner crystallization in the honeycomb optical lattice. *Phys. Rev. Lett.* **99,** 070401 (2007).
16. Browaeys, A. *et al*. Transport of atoms in a quantum conveyor belt. *Phys. Rev. A* **72,** 053605 (2005).
17. Köhl, M., Moritz, H., Stöferle, T., Günter, K. & Esslinger, T. Fermionic atoms in a three dimensional optical lattice: Observing Fermi surfaces, dynamics, and interactions. *Phys. Rev. Lett.* **94,** 080403 (2005).
18. Müller, T., Fölling, S., Widera, A. & Bloch, I. State preparation and dynamics of ultracold atoms in higher lattice orbitals. *Phys. Rev. Lett.* **99,** 200405 (2007).
19. Marston, J. B. & Affleck, I. Large-n limit of the Hubbard–Heisenberg model. *Phys. Rev. B* **39,** 11538–11558 (1989).
20. Hemmerich, A. & Hänsch, T. Radiation pressure vortices in two crossed standing waves. *Phys. Rev. Lett* **68,** 1492–1495 (1992).
21. Anderlini, M. *et al*. Controlled exchange interaction between pairs of neutral atoms in an optical lattice. *Nature* **448,** 452–456 (2007).
22. Klinner, J., Wolke, M. & Hemmerich, A. Increased efficiency of rf-evaporative cooling by utilizing gravity. *Phys. Rev. A* **81,** 043414 (2010).
23. Hemmerich, A. & Morais Smith, C. Excitation of a d-density wave in an optical lattice with driven tunneling. *Phys. Rev. Lett.* **99,** 113002 (2007).
24. Lim, L-K., Morais Smith, C. & Hemmerich, A. Staggered-vortex superfluid of ultracold bosons in an optical lattice. *Phys. Rev. Lett.* **100,** 130402 (2008).
25. Lim, L-K., Hemmerich, A. & Morais Smith, C. Staggered-Vortex superfluid of ultracold bosons in an optical lattice. *Phys. Rev. A* **81,** 023404 (2010).
26. Stojanović, V. M., Wu, C., Liu, W. V. & Das Sarma, S. Incommensurate superfluidity of bosons in a double-well optical lattice. *Phys. Rev. Lett.* **101,** 125301 (2008).
27. Partridge, G. B., Li, W., Kamar, R. I., Liao, Y. & Hulet, R. G. Pairing and phase separation in a polarized Fermi gas. *Science* **311,** 503–505 (2006).
28. Zwierlein, M. W., Schunck, C. H., Schirotzek, A. & Ketterle, W. Direct observation of the superfluid phase transition in ultracold Fermi gases. *Nature* **442,** 54–58 (2006).
29. Loois, C. C., Barkema, G. T. & Morais Smith, C. Monte Carlo studies of extensions of the Blume–Emery–Griffiths model. *Phys. Rev. B* **78,** 184519 (2008).
30. Bakr, W. S, Gillen, J. I., Peng, A., Fölling, S. & Greiner, M. A quantum gas microscope for detecting single atoms in a Hubbard-regime optical lattice. *Nature* **462,** 74–77 (2009).

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Additional information

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